

# Rectangle-constant closure: the theorem-grade rectangle prefactor, the incidence route, and the admissibility closure condition (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

The v2.0 main-chain note left two items open: the rectangle prefactor  $c_R$  (measured, not derived) and the extreme- $n$  rich-carrier corner. This companion note (i) archives the OPERATOR-SUPPLIED derivation (review verdict #9) that closes  $c_R$  at theorem grade, (ii) contributes a NEW incidence-geometric route that lowers the unconditional exponent from  $n^{5/2}$  to  $n^{28/13}$  polylog, and (iii) reduces full STEP-5B closure to one named, quantitatively explicit hypothesis (H-ADM) plus the  $\kappa$ -balance scope (H-KBAL). It claims NEITHER unconditional closure NOR any tier action; tier decisions remain with the operator.

## 2. Theorem-grade rectangle constant (operator derivation, verdict #9)

**Proposition 1** (Rigorous  $\sqrt{n}$  energy bound; operator-supplied). *Let  $N = 2n = |\hat{Q}|$  and let  $p_C$  be the ordered antipodal pair count of carrier  $C$ . Three distinct points lie on at most one circle, so*

$$\sum_{C: k_C \geq 3} p_C^3 \leq \sum_{C: k_C \geq 3} k_C^3 \leq 27 \binom{N}{3} < \frac{9}{2} N^3, \quad \sum_{C: p_C \leq 2} p_C^3 \leq 4 \sum_C p_C \leq 4N^2,$$

hence  $\sum_C p_C^3 \leq 7N^3$  conservatively. With  $\sum_C p_C \leq N^2$  and Cauchy-Schwarz interpolation,

$$\sum_C p_C^2 \leq \left( \sum_C p_C \right)^{1/2} \left( \sum_C p_C^3 \right)^{1/2} \leq \sqrt{7} N^{5/2}.$$

For  $\kappa$ -balanced patterns ( $A_{\max}^2 \leq \kappa^2 I/n = 2\kappa^2 I/N$ ),  $\Psi_C \leq p_C A_{\max}^2$  gives

$$\sum_{t \neq 0} w_t^2 \leq \lambda^2 I^2 \left( 8 + 4\sqrt{14} \kappa^4 \sqrt{n} \right), \quad c_R^{\text{rigorous}} = 4\sqrt{14} \simeq 14.97.$$

The v2.0 calibrated corollary is hereby THEOREM-grade; the certified region at the anchor budget ( $K_{\text{budget}} = 5972$  at  $I = 4 \times 10^{-4}$ ) is  $n \lesssim 1.59 \times 10^5$  ( $3.77 \times 10^3$  at  $10^{-3}$ ;  $2.02 \times 10^2$  at  $2 \times 10^{-3}$ ). Machine checks:  $\sum p^3 \leq 7N^3$  and the interpolation hold on all test configurations.

### 3. New incidence route: exponent 28/13

**Theorem 1** (Stereographic transfer). *A stereographic projection from a pole not on any carrier maps the sphere circles bijectively to circles (or lines) in the plane, preserving all point-circle incidences. (Machine check: projected ring circles are exact plane circles, residuals  $\sim 10^{-30}$ .)*

**Theorem 2** (Incidence upgrade). *Let  $m_r$  be the number of carriers with  $p_C \geq r$ . The pair partition gives the cap  $m_r \leq 2N^2/r$ . In the plane, the Aronov-Sharir-type point-circle incidence bound  $\mathcal{I} \leq C(N^{2/3}m^{2/3} + N^{6/11}m^{9/11}L + N + m)$  ( $L$  polylogarithmic) yields the rich-circle bounds  $m_r \leq C(N^2/r^3 + N^3L^{11/2}/r^{11/2} + N/r)$ . Optimizing the dyadic sum  $\sum_C p_C^2 = \sum_r m_r r^2$  against the crossover of the cap  $2N^2/r$  with  $N^3L/r^{11/2}$  (at  $r_1 = (NL)^{2/13}$ ) gives*

$$\sum_C p_C^2 = O(N^{28/13} \text{polylog } N), \quad \frac{28}{13} \simeq 2.154 < \frac{5}{2},$$

and consequently, for  $\kappa$ -balanced patterns,

$$K(n) \leq 8 + c\kappa^4 n^{2/13} \text{polylog } n.$$

At the anchor budget this certifies all  $n \lesssim 7.9 \times 10^{16}$  — eleven orders of magnitude beyond the  $\sqrt{n}$  route.

*Remark 1* (Sharp conjecture, pre-registered). Measured growth exponents of  $\sum_C p_C^2$  on the worst-known families are  $n^{2.06}$  (rings),  $n^{2.04}$  (random),  $n^{2.08}$  (coaxial): all consistent with the SHARP conjecture  $\sum_C p_C^2 = O(N^2 \text{polylog})$ , under which  $K(n)$  grows only polylogarithmically and the certified region exceeds any physically meaningful mode count for every admissible intensity. The conjecture is pre-registered with falsification gate: any family with measured exponent  $\geq 2.3$  refutes it.

### 4. Closure condition (named hypotheses)

**Theorem 3** (Conditional STEP-5B closure). *Under the named hypotheses*

$$\mathbf{H-KBAL}: A_{\max}^2 \leq \kappa^2 I/n \text{ (balanced amplitudes, } \kappa \text{ explicit)}, \quad \mathbf{H-ADM}: n \leq n_{\text{adm}},$$

STEP-5B holds for every admissible pattern provided

$$8 + c\kappa^4 n_{\text{adm}}^{2/13} \text{polylog} < K_{\text{budget}}(I),$$

which at the production anchor is satisfied by ANY  $n_{\text{adm}} < 7.9 \times 10^{16}$ . The operator-noted packing form  $n \leq C_{\text{pack}}(q_0/\Delta q_{\min})^2$  converts this into the mode-separation condition  $\Delta q_{\min}/q_0 \gtrsim 10^{-8}$  — i.e. shell directions need only be distinguishable at the  $10^{-8}$  angular scale, far weaker than any physical coherence or UV constraint. The unbalanced-amplitude extension (amplitude-dyadic decomposition over  $\sim \log n$  classes) is registered as the remaining technical lift of H-KBAL.

### 5. Numerical verification

Script [codes/vacuum/beyond\\_layer\\_gershgorin\\_bound.py](#) (v1.9.0; 166/166 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

| Quantity                      | Value                                                    | Check                               |
|-------------------------------|----------------------------------------------------------|-------------------------------------|
| operator $p^3$ bound          | $\sum p^3 \leq 7N^3$ on rand/ring/coax                   | verdict-#9 constants verified       |
| interpolation                 | $\sum p^2 \leq \sqrt{\sum p} \sum p^3$                   | all configs                         |
| $c_R$ rigorous                | $4\sqrt{14} = 14.9666$                                   | theorem-grade                       |
| theorem region ( $\sqrt{n}$ ) | $1.59 \times 10^5 / 3.77 \times 10^3 / 2.02 \times 10^2$ | matches operator $1.58 \times 10^5$ |
| stereographic transfer        | circle residuals $10^{-30}$                              | incidences preserved                |
| incidence region              | $7.9 \times 10^{16}$ at anchor                           | exponent 28/13                      |
| growth exponents              | 2.06/2.04/2.08 (ring/rand/coax)                          | sharp- $O(n^2)$ evidence            |

Quantitative sanity check (mandatory): the operator’s region figure  $1.58 \times 10^5$  is reproduced independently ( $1.59 \times 10^5$ , rounding);  $28/13 < 5/2$  strictly; the  $n$ -reach ratio between routes ( $7.9 \times 10^{16} / 1.59 \times 10^5 \approx 5 \times 10^{11}$ ) equals  $((K_b - 8)/c)^{13/2-2}$ -scaling as dimensional consistency requires.

## 6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- ( $\alpha$ ) “*The Aronov–Sharir incidence bound is for plane circles; carriers live on the sphere.*” DISMISSED: stereographic projection from a generic pole is a bijection sphere-circles  $\rightarrow$  plane-circles/lines preserving incidences (machine-verified exact); the planar theorem applies verbatim to the projected family.
- ( $\beta$ ) “*The polylog factors and the constant  $c$  in the 28/13 route are not pinned.*” VALID-with-mitigation: the EXPONENT is unconditional; the polylog/constant affect the  $n$ -reach only through a 13/2 power, and even a constant as large as  $10^3$  leaves  $n_{\text{reach}} > 10^{10}$ . Pinning the constant is registered as the companion sub-task; all quoted regions use the conservative form.
- ( $\gamma$ ) “*H-KBAL ( $\kappa$ -balance) may exclude physical competitors.*” VALID-with-mitigation: heavily unbalanced patterns are dominated by their heavy modes and fall into the small- $n_{\text{eff}}$  certified region (the heavy sub-pattern carries the energy); the rigorous amplitude-dyadic lift is registered. No admissible competitor class is known to violate H-KBAL at fixed  $I$ .
- ( $\delta$ ) “*The sharp  $O(n^2)$  claim rests on three families.*” DISMISSED as stated: the  $O(n^2)$  statement is registered as a CONJECTURE with a pre-registered falsification gate (exponent  $\geq 2.3$  on any family), not as a theorem; no closure quote depends on it.

## 7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / AddA (serving STEP-5B)

Precise statement: (i) THEOREM (operator-derived, verdict #9):  $K(n) \leq 8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$ ; region  $1.59e5$  at anchor. (ii) THEOREM (incidence route):  $\sum_C p_C^2 = O(N^{\{28/13\}} \text{polylog})$  via stereographic transfer + Aronov-Sharir; region  $7.9e16$  at anchor. (iii) CONDITIONAL CLOSURE: STEP-5B holds under  $\{H\text{-KBAL}, H\text{-ADM}\}$  with  $n_{\text{adm}} < 7.9e16$  (mode separation  $> \sim 1e-8$ ). (iv) Sharp  $O(n^2)$  pre-registered conjecture (measured exponents 2.04-2.08).

Scope:  $\kappa$ -balanced amplitudes (H-KBAL); second-

|                         |                                                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | cumulant Hessian scope of the main chain;<br>unbalanced lift registered.                                                                                                                                                                                                                |
| Dependencies:           | B5 main chain v2.0 (carrier partition, rectangle reformulation, triple count); A1-KERNEL-CONV; margins from Proposition A (B2 chain).                                                                                                                                                   |
| Evidence grade:         | ANALYTIC (both routes) + EXECUTED (verification) + CITED (planar incidence literature)                                                                                                                                                                                                  |
| Reproduction command:   | python codes/vacuum/beyond_layer_gershgorin_bound.py                                                                                                                                                                                                                                    |
| Expected output:        | claims 166/166 PASS; result.json under<br>claims/B5-BEYOND-LAYER-BOUND/runs/<br>260605-gershgorin-reduction/                                                                                                                                                                            |
| Falsification gate:     | Any family with sum $p^2$ growth exponent $\geq 2.3$<br>(refutes sharp conjecture, not the theorems);<br>any kappa-balanced pattern with $n < 1.59e5$<br>violating the margin (refutes the theorem chain).                                                                              |
| Tier before / after:    | T4 / T4 (TIER PROPOSAL for operator: T5, or T6<br>CONDITIONAL on {H-KBAL, H-ADM} - decision is<br>the operator's)                                                                                                                                                                       |
| No-overclaim statement: | Unconditional class-wide closure (all $n$ , all<br>amplitude profiles) is NOT claimed: it requires<br>the sharp $O(n^2)$ conjecture or H-ADM. The<br>incidence constant/polylog is conservative, not<br>pinned. No tier action on B1/B2; Reading-H<br>stays T5 pending operator review. |
| Next required action:   | Operator tier review of the closure-conditional<br>status; amplitude-dyadic H-KBAL lift; pin the<br>incidence constant; (optional) sharp $O(n^2)$ .                                                                                                                                     |